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#include <stdio.h>

#include <stdlib.h>

// Node structure

struct Node {

    int data;

    struct Node* next;

};

// Function to create a new node

struct Node* createNode(int data) {

    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    newNode->data = data;

    newNode->next = NULL;

    return newNode;

}

// ----- STACK OPERATIONS -----

// Push operation (insert at beginning)

void push(struct Node** top, int data) {

    struct Node* newNode = createNode(data);

    newNode->next = *top;

    *top = newNode;

    printf("Pushed %d onto stack\n", data);

}

// Pop operation (delete from beginning)

int pop(struct Node** top) {

    if (*top == NULL) {

        printf("Stack Underflow!\n");

        return -1;

    }

    struct Node* temp = *top;

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    int popped = temp->data;

    *top = temp->next;

    free(temp);

    return popped;
}

// Display stack
void displayStack(struct Node* top) {

    printf("Stack: ");

    while (top != NULL) {

        printf("%d -> ", top->data);

        top = top->next;

    }

    printf("NULL\n");

}

// ----- QUEUE OPERATIONS -----

// Enqueue operation (insert at end)
void enqueue(struct Node** front, struct Node** rear, int data) {

    struct Node* newNode = createNode(data);

    if (*rear == NULL) { // empty queue

        *front = *rear = newNode;

    } else {

        (*rear)->next = newNode;

        *rear = newNode;

    }

    printf("Enqueued %d into queue\n", data);

}

// Dequeue operation (delete from front)
int dequeue(struct Node** front, struct Node** rear) {

    if (*front == NULL) {

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        printf("Queue Underflow!\n");
        return -1;
    }
    struct Node* temp = *front;
    int dequeued = temp->data;
    *front = temp->next;
    if (*front == NULL) { // queue becomes empty
        *rear = NULL;
    }
    free(temp);
    return dequeued;
}

// Display queue
void displayQueue(struct Node* front) {
    printf("Queue: ");
    while (front != NULL) {
        printf("%d -> ", front->data);
        front = front->next;
    }
    printf("NULL\n");
}

// ----- MAIN FUNCTION -----

int main() {
    // Stack simulation

    struct Node* stackTop = NULL;

    push(&stackTop, 10);
    push(&stackTop, 20);
    push(&stackTop, 30);
    displayStack(stackTop);
}

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printf("Popped: %d\n", pop(&stackTop));
displayStack(stackTop);

// Queue simulation

struct Node* front = NULL;
struct Node* rear = NULL;

enqueue(&front, &rear, 100);
enqueue(&front, &rear, 200);
enqueue(&front, &rear, 300);
displayQueue(front);

printf("Dequeued: %d\n", dequeue (&front, &rear));
displayQueue(front);

return 0;
}

```

OUTPUT:

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Pushed 10 onto stack
Pushed 20 onto stack
Pushed 30 onto stack
Stack: 30 -> 20 -> 10 -> NULL
Popped: 30
Stack: 20 -> 10 -> NULL
Enqueued 100 into queue
Enqueued 200 into queue
Enqueued 300 into queue
Queue: 100 -> 200 -> 300 -> NULL
Dequeued: 100
Queue: 200 -> 300 -> NULL

Process returned 0 (0x0)   execution time : 1.593 s
Press any key to continue.

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